Kernix 0.10.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

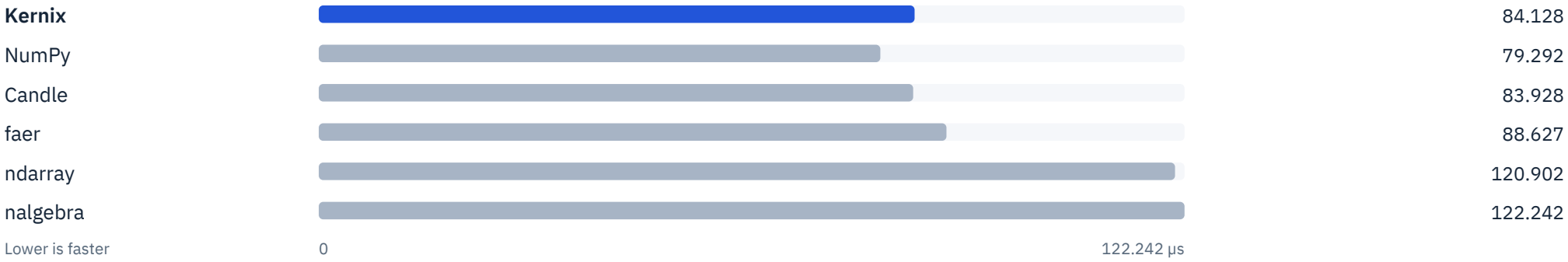
31 workloads

108 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

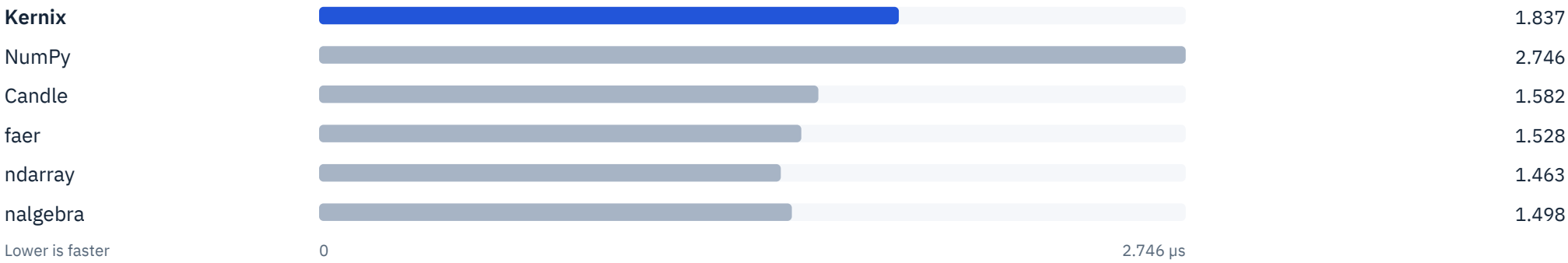
Kernix 0.10.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

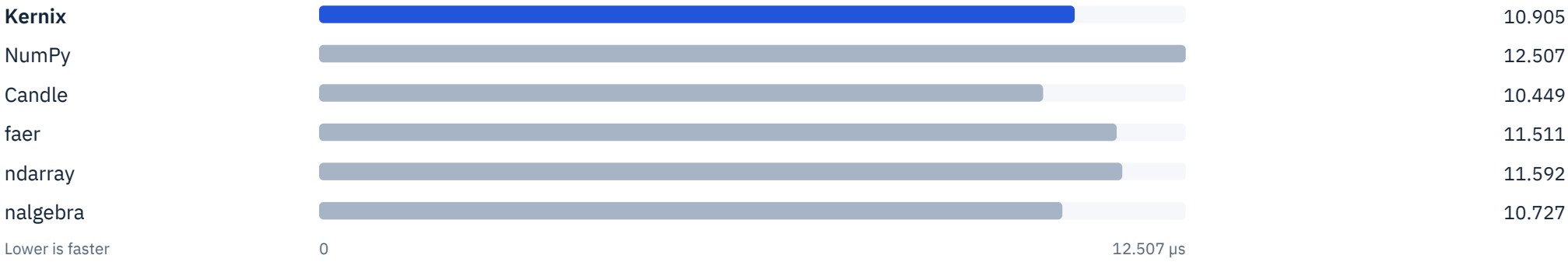
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



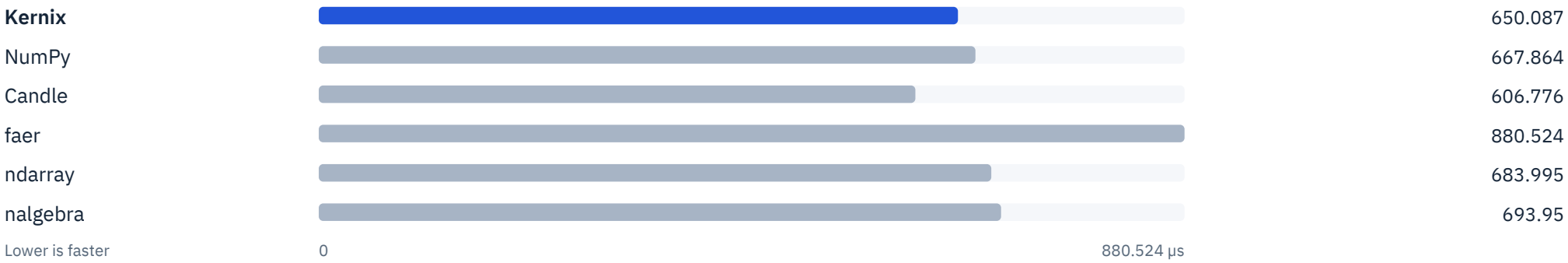
Kernix 0.10.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

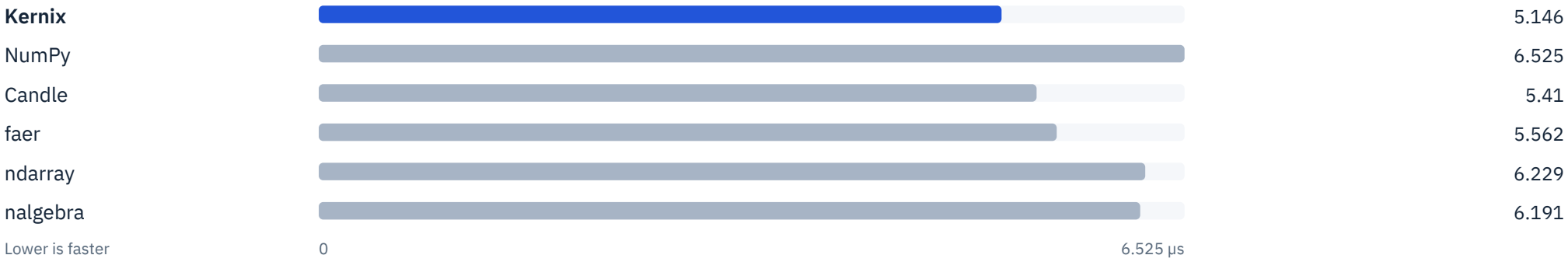
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



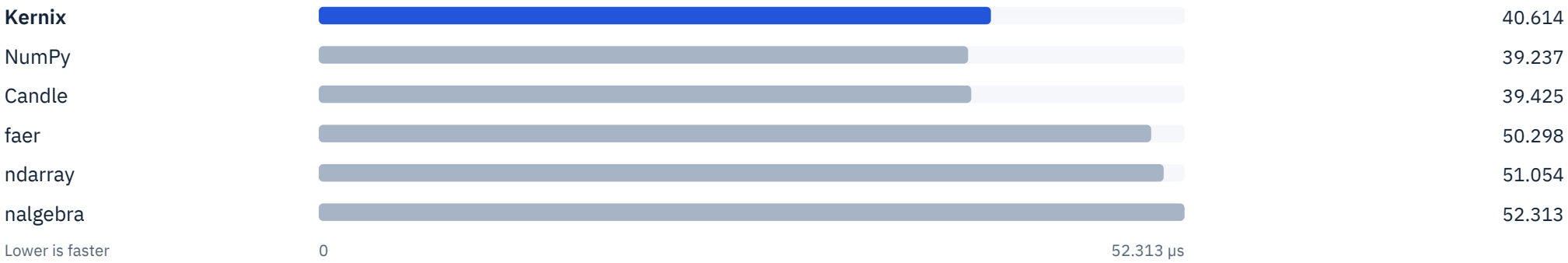
Kernix 0.10.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

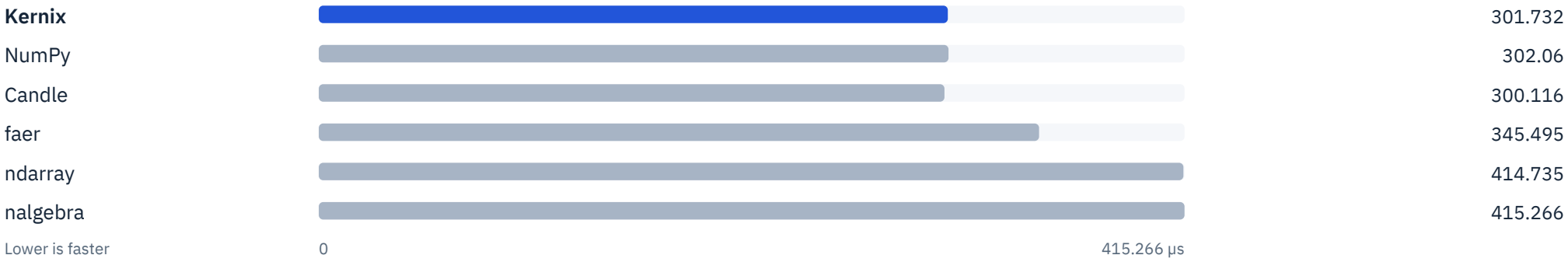
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



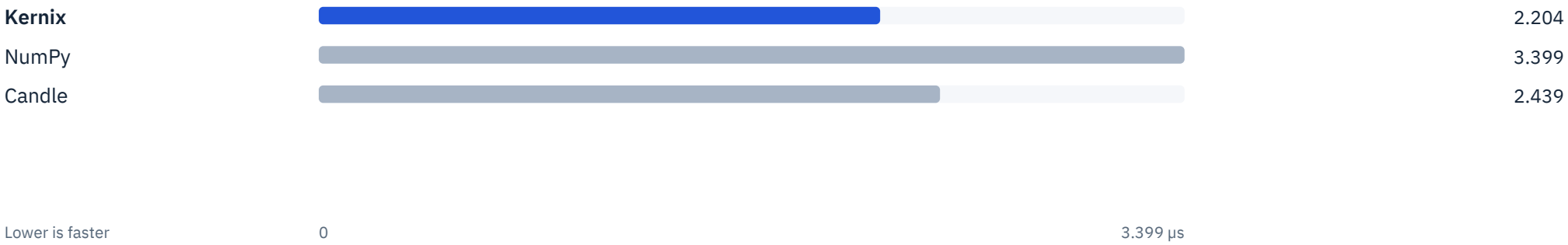
Kernix 0.10.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.10.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

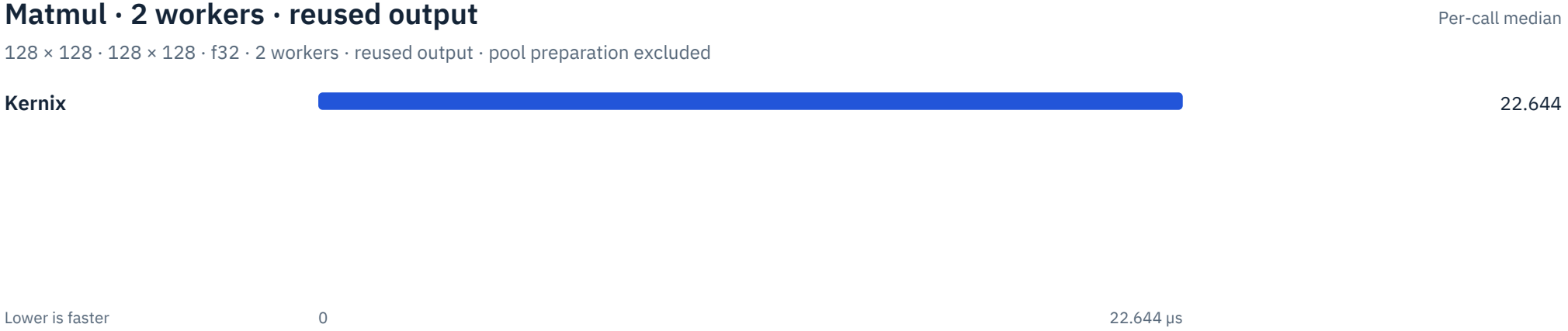
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

Per-call median

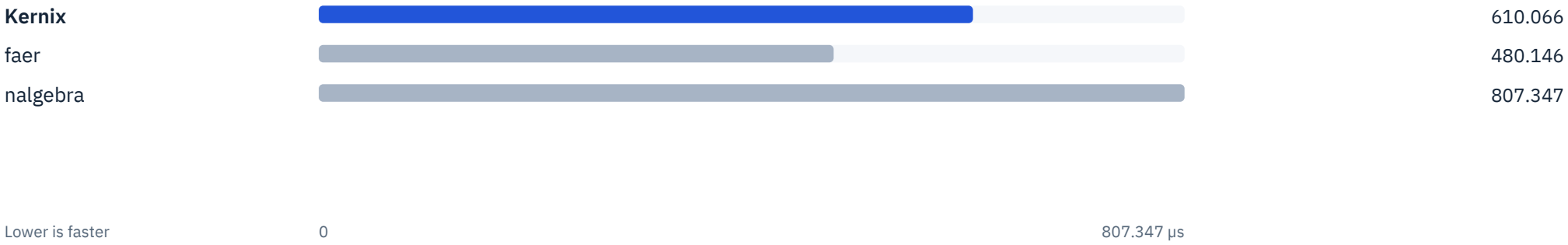
96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

Per-call median

256 × 64 · f64 · 1 worker · allocated output



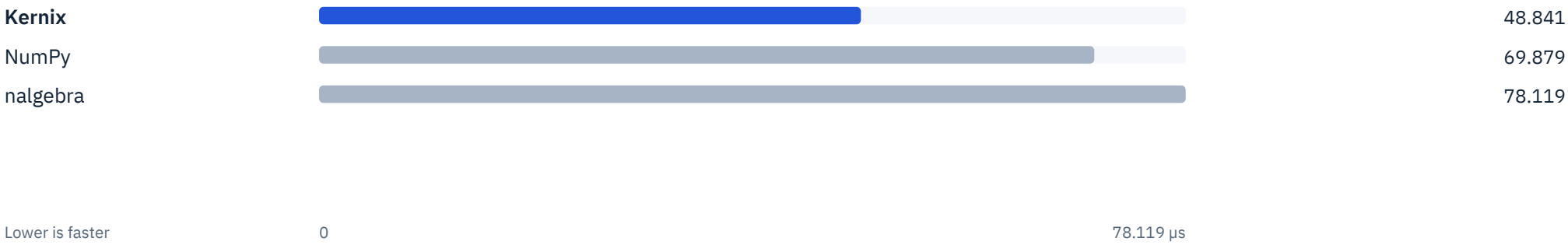
Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

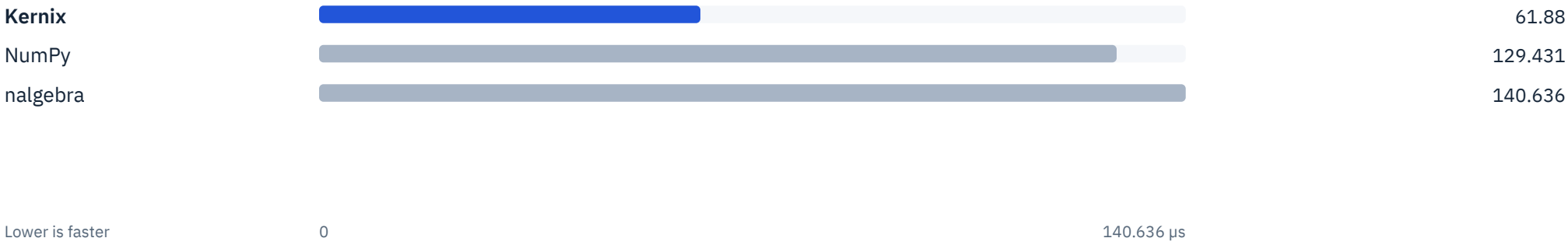
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



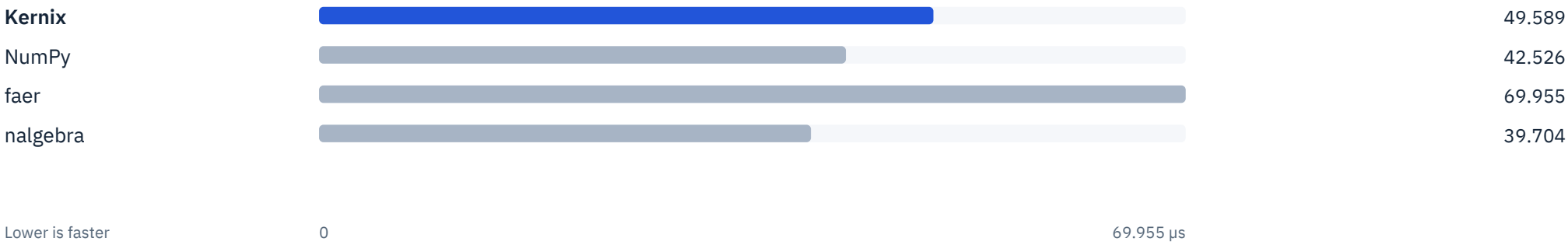
Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

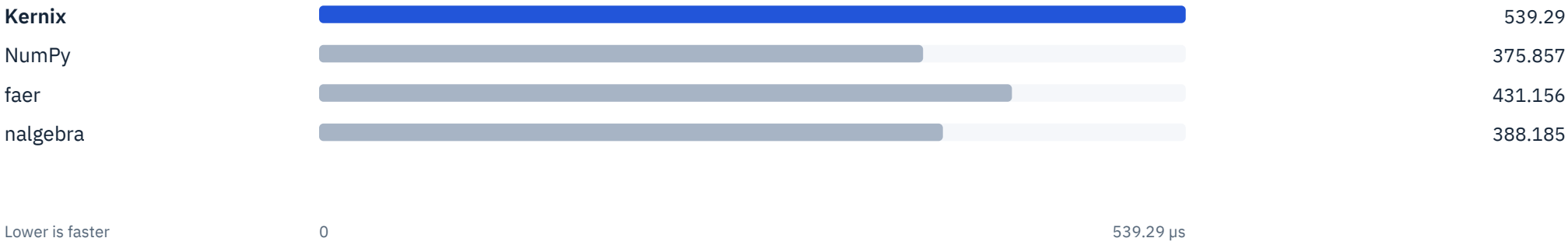
Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



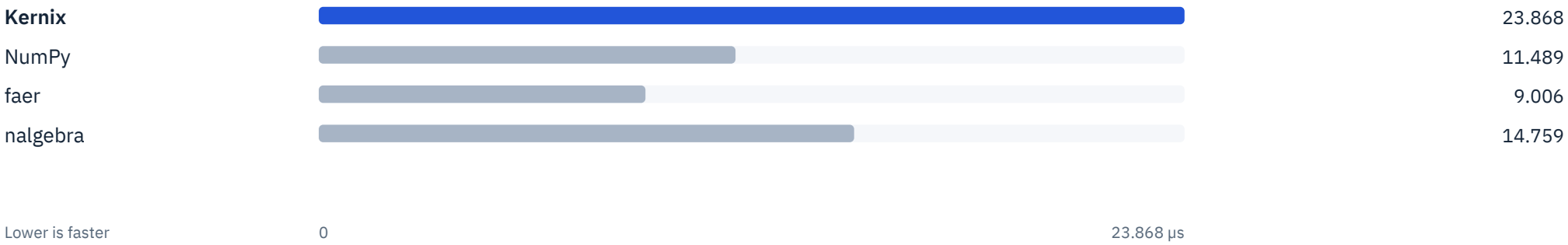
Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

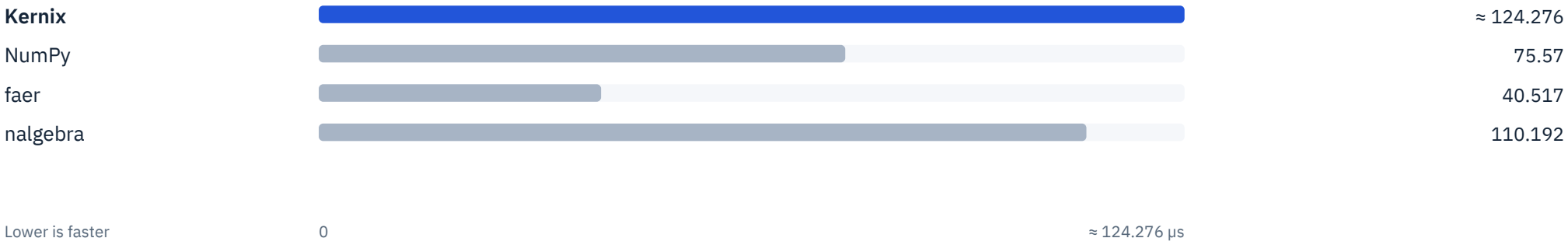
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



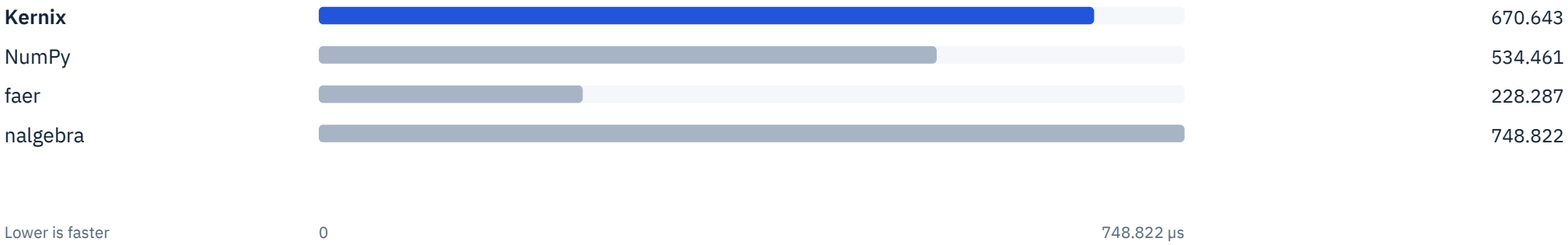
Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



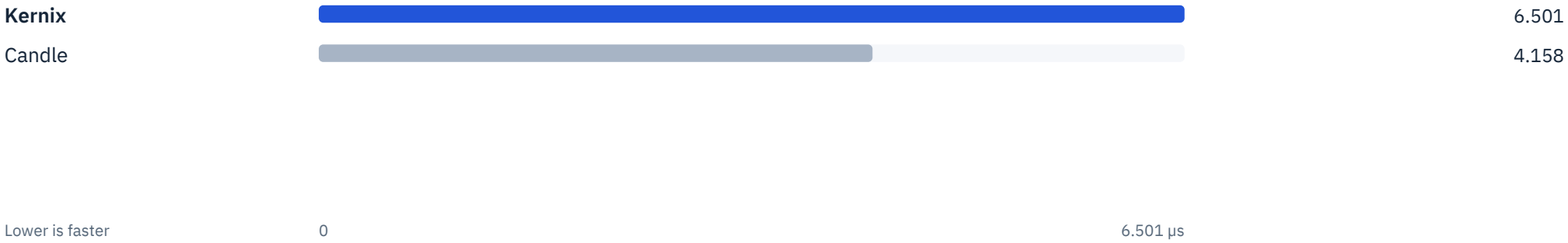
Kernix 0.10.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



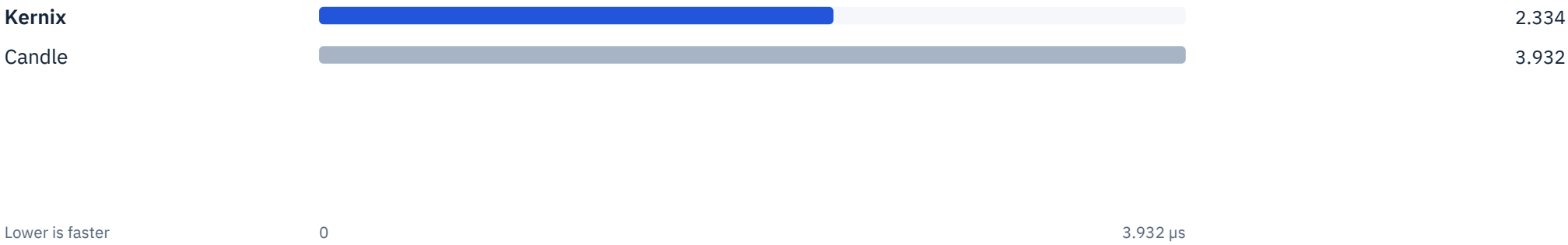
Kernix 0.10.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



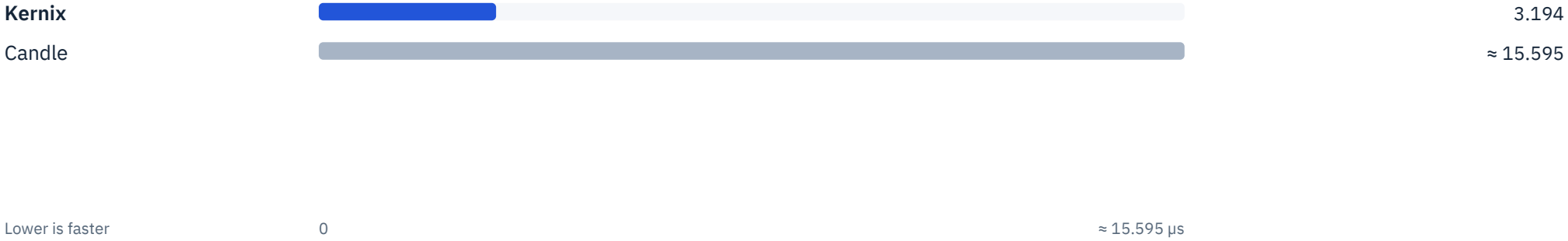
Kernix 0.10.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.10.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



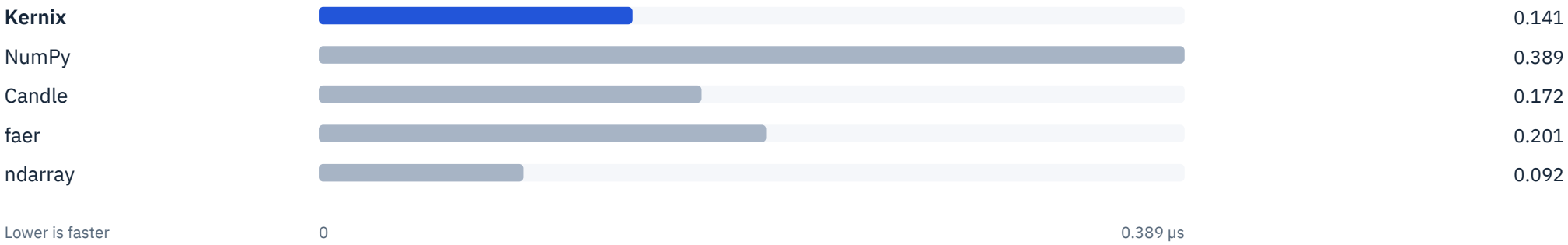
Kernix 0.10.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9601934957454249	0.9392138063279003 – 1.005085111813559

Minimum-norm least squares

32 × 64, rank 32, 4 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7588965246897702	0.7181894083969466 – 0.7686042740278929

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.05853394018885557	0.0575878995545003 – 0.061997534457761586

Shared-variable backward

sum(X * X), 64 × 32; prepared graph · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9687731241675299	0.9081661486295564 – 1.005344231373782

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	0.939817091159406	95% INTERVAL	0.4675953414021466 – 1.4206664022213407

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	0.7737991100292854	95% INTERVAL	0.768674522566667 – 0.8446403359519241

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7363103488972145	0.7254634944389963 – 0.7608528926076653

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7687074829931972	0.4377720919575346 – 0.8030099989691785

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0172824791418356	0.9823877983479045 – 1.0442190966043439

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9942393049390877	0.9576774193548387 – 1.0324791345424285

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Thin QR

96 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0127366207057724	0.9813273721187985 – 1.0610946896340843

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0036583839038473	0.970302782264253 – 1.0459129820712867

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	**1.0003607237430008**	95% INTERVAL	0.9832872672183858 – 1.022784057586786

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1528	1.528
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1582	1.582
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2746	2.746
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1837	1.837
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1463	1.463
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1498	1.498
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11592	11.592
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	11511	11.511
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	10449	10.449
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	10905	10.905

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	12507	12.507
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	10727	10.727
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	84128	84.128
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	88627	88.627
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	120902	120.902
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	122242	122.242
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	83928	83.928
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	79292	79.292
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	650087	650.087
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	667864	667.864

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	606776	606.776
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	880524	880.524
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	693950	693.95
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	683995	683.995
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6191	6.191
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	5410	5.41
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5146	5.146
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5562	5.562
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6229	6.229
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6525	6.525

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	50298	50.298
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	39237	39.237
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	39425	39.425
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	51054	51.054
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	40614	40.614
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	52313	52.313
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	345495	345.495
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	301732	301.732
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	300116	300.116
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	415266	415.266

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	414735	414.735
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	302060	302.06
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2439	2.439
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3399	3.399
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2204	2.204
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	9832	9.832
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	81816	81.816
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	307403	307.403
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	22644	22.644
Column-pivoted QR / Kernix 96 × 32 · f64	58549	58.549

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	77385	77.385
Column-pivoted QR / nalgebra 96 × 32 · f64	81493	81.493
Column-pivoted QR / Kernix 256 × 64 · f64	610066	610.066
Column-pivoted QR / faer 256 × 64 · f64	480146	480.146
Column-pivoted QR / nalgebra 256 × 64 · f64	807347	807.347
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	48841	48.841
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	78119	78.119
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	69879	69.879
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	140636	140.636
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	61880	61.88

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	129431	129.431
Thin QR / nalgebra 96 × 32 · f64	39704	39.704
Thin QR / NumPy 96 × 32 · f64	42526	42.526
Thin QR / Kernix 96 × 32 · f64	49589	49.589
Thin QR / faer 96 × 32 · f64	69955	69.955
Thin QR / faer 256 × 64 · f64	431156	431.156
Thin QR / Kernix 256 × 64 · f64	539290	539.29
Thin QR / NumPy 256 × 64 · f64	375857	375.857
Thin QR / nalgebra 256 × 64 · f64	388185	388.185
Cholesky factorization / NumPy 64 × 64 · f64	11489	11.489

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	23868	23.868
Cholesky factorization / faer 64 × 64 · f64	9006	9.006
Cholesky factorization / nalgebra 64 × 64 · f64	14759	14.759
Cholesky factorization / nalgebra 128 × 128 · f64	110192	110.192
Cholesky factorization / NumPy 128 × 128 · f64	75570	75.57
Cholesky factorization / faer 128 × 128 · f64	40517	40.517
Cholesky factorization / Kernix 128 × 128 · f64	124275.99999999999	124.27599999999999
Cholesky factorization / nalgebra 256 × 256 · f64	748822	748.822
Cholesky factorization / faer 256 × 256 · f64	228287	228.287
Cholesky factorization / Kernix 256 × 256 · f64	670643	670.643

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	534461	534.461
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4158	4.158
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6501	6.501
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4581	4.581
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2582	2.582
Model inference / Candle batched normalized/sliced MLP · f64	3932	3.932
Model inference / Kernix batched normalized/sliced MLP · f64	2334	2.334
Model recording + forward / Candle batched normalized/sliced MLP · f64	5592	5.592
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2774	2.774
Model backward / Kernix prepared graph → fresh gradients · f64	3194	3.194

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Candle prepared graph → fresh gradients · f64	15594.999999999998	15.594999999999998
Parameter update / Kernix transactional SGD, fixed gradients · f64	584	0.584
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	39133	39.133
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	8329	8.329
Row softmax / Candle 256 × 64 · f64	89845	89.845
Row softmax / ndarray 256 × 64 · f64	85585	85.585
Row softmax / Kernix 256 × 64 · f64	93964	93.964
Row softmax / NumPy 256 × 64 · f64	110693	110.693
Create zeros / Candle 64 × 64 · f32	172	0.172
Create zeros / faer 64 × 64 · f32	201	0.201

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	141	0.141
Create zeros / ndarray 64 × 64 · f32	92	0.092
Create zeros / NumPy 64 × 64 · f32	389	0.389
Create ones / Kernix 64 × 64 · f32	196	0.196
Create ones / Candle 64 × 64 · f32	231	0.231
Create ones / NumPy 64 × 64 · f32	1406	1.406
Create ones / ndarray 64 × 64 · f32	154	0.154
Create ones / faer 64 × 64 · f32	193	0.193